

The Move From Inpatient To Outpatient Settings And The Use of Artificial Intelligence in The US Healthcare Sector: Lessons For Other Countries

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Abstract. This paper explores the possibility of applying artificial intelligence (AI) tools to the US healthcare sector for achieving reductions in healthcare cost and enhancing efficiency and better outcome for patients. The paper identifies specific areas of healthcare, particularly in the context of outpatient sector, where AI tools can be applied. The paper also proposes how the best AI practices in the US healthcare can be adopted and adapted by other countries and scaled up. The paper concludes with a discussion of the challenges that may stymie the application of AI to healthcare and its diffusion to other countries.

Keywords: Artificial Intelligence (AI), healthcare, US healthcare, outpatient services.

1. Introduction

The United States (US) represents the world's largest and most dynamic healthcare market, characterized by an unmatched concentration of healthcare infrastructure, resources, and expertise. This ecosystem encompasses a vast network of consumers (e.g., patients), research labs, research universities, pharmaceutical companies, drug manufacturers, medical equipment manufacturers, and healthcare facilities. Such a comprehensive infrastructure, combined with substantial healthcare spending and risk-taking spirit of American people (Hofstede, 2010), creates an ideal environment for technological innovation and novel experimentation in US healthcare practices. Despite spending more on per capita healthcare than anywhere else in the world, the US healthcare system is also plagued by many problems. The rising healthcare costs, which reached US\$ 4.7 trillion in 2023 (McGonough *et al.*, 2024), is also unsustainable in the long run. Against this backdrop, it can be argued that the US healthcare industry is ripe for a major shakeup and Artificial Intelligence (AI), which has emerged as a transformative technology (Al-Kuwaiti *et al.*, 2023), offers promising solutions to various healthcare challenges including prognosis, diagnosis, and analysis of diseases (Kumar and Divya, 2024) as well as significantly reducing healthcare costs (Khanna *et al.*, 2022).

The US healthcare system's role as a global pioneer means that healthcare practices in the US often get adopted by other countries, influencing healthcare delivery worldwide. This paper is based on the premise that various AI tools and technologies, initially developed and tested in the US healthcare market, can be quickly adapted to address healthcare challenges in different national contexts, taking into account their varying levels- including inadequacies- of resources, infrastructure, and healthcare systems.

This paper examines the potential applications of AI to the US healthcare sector, with a specific focus on *outpatient care*, and explores how these innovations might be adapted and implemented in other countries, including Jordan. However, it is important to acknowledge that AI is a disruptive technology (Bajwa *et al.*, 2024) and the nascent nature of AI applications in healthcare leaves a lot of unknowns which can only be clarified with passage of time. Given the limited availability of comprehensive empirical studies and well-documented case studies, many of the discussions presented in this paper are aspirational to some extent. Nevertheless, this discussion is particularly relevant and timely as healthcare systems globally are beset with challenges of accessibility, affordability, equity, and quality of care.

2. Growth of Outpatient Service in the US

In an ideal scenario, quality healthcare should be affordable for and accessible by most of the population of a given country. Further, a key feature has to be transparency and accountability. Towards that end, in the US, there has been a steady shift of cases from inpatient (i.e., overnight stay in hospital or healthcare facility) to outpatient (i.e., returning home the same day of receiving the treatment/procedure) settings (Lao, McCaig and Hing, 2016). This has been made possible by the improvement of technology and desire to reduce medical costs.

For the past two decades, the trend of performing surgeries outside of inpatient settings has been driven by surgical technique advancement as well as availability of data suggesting that patients recover better at home with less risk of infection. Moreover, outpatient service also leads to significant cost savings and better personalized care.

Surgeries in outpatient settings have become routine in cases like cataract removal, colonoscopies, endoscopies, and even many orthopedics procedures. It is not uncommon to have total knee, hip and shoulder replacements done in ambulatory settings and patients getting released the same day with monitoring devices. Cardiovascular surgeries is also emerging as the new frontier in outpatient surgery.

3. Application of AI to Healthcare

AI, an emergent technology, seems poised to reshape the healthcare industry in the US as well as in other countries. Recent advances in AI is beginning to affect significantly and positively the quality and cost of healthcare at every level with lessons for both developed and emerging countries. Some of the areas where AI can be applied with relative ease and quickly scaled up are as follows:

Clinical Decision Support: AI can help with various types of clinical decision support such as diagnostic assistance, robotic surgery, planning and execution of treatment, checking interactions of various medicines, etc. Such AI augmented support system can also lead to reduced human error and enhance the efficiency of the healthcare delivery.

Early Detection and Better Diagnosis: AI tools can play a pivotal role in detecting diseases early and thereby enhancing the possibility of successful treatment and outcome for the patients. Such early detections can lead to better preventive care. AI can also help with pattern recognition of several ailments such as autism, dementia, and various degenerative diseases about which our understanding is limited. AI powered imaging software can also help physicians to make more accurate diagnosis and plan treatment strategies. AI helps the development of 3D images which make surgeries less risky and can inform and educate the patients of their condition. It also reduces the possibility of medical malpractice lawsuits and consequently lowers insurance costs for medical practice.

Enhanced Administrative Efficiency: Application of AI tools can lead to better work flow management. AI tools can help with communicating with the patients, appointment and follow up reminders, compliance with relevant statutes and reporting to relevant government agencies, communicating with insurance providers and billing optimizations, etc.

Staffing Challenges: Many countries experience shortages of qualified healthcare professionals and staff and this is where AI can be a positive tool for training staff and monitoring patient care. A number of intelligent virtual care platforms are already in use to orient and train staff at a fraction of current onboarding and training costs. These training tools can be accessed round-the-clock by nurses and other healthcare staff. Some of these platforms deploy AI-powered virtual nursing solutions and drive measurable outcomes with support from care experts.

Accelerated Drug Development: Developing new drugs can be time consuming and prohibitively expensive. AI can significantly reduce both the time and development costs and contribute to analysis at molecular level and optimization of clinical trial.

Continuous Patient Monitoring: Aided by AI tools, virtual patient monitoring (VPM) can be achieved at significantly reduced costs in both inpatient and outpatient settings. AI assisted VPM is making it possible for clinicians to access patients in remote settings as well as in detection and prevention of adverse events like falls or self-harm by mental patients. A single clinician can now monitor multiple patients with the use of AI driven monitors and react timely to acuity alerts with deterioration algorithms. Wearable devices like watches driven by AI can monitor information and send alerts (when necessary) about heart rates, stress management, and sleep patterns.

Greater Health Access: Various AI tools can help with ensuring better health access by people who currently have limited access to healthcare. AI tools can significantly enhance the scope and reach of telemedicine, especially in developing countries. It can also help bridge the gap between developed and developing countries as well as reduce the disparity between urban and rural healthcare facilities.

Patient Education and Virtual Assistants: AI powered computer programs are already being extensively used for making appointments, providing information on medical conditions, and sending reminders for taking medication. This effectively reduces the need for in-person visits. Such routine tasks can lead to significant cost reduction.

The following Table summarizes these potential application of AI to the healthcare sector.

Table 1: Application of AI to Healthcare

Specific Areas of Healthcare	Potential AI Applications
Clinical Decision Support	• Diagnostic Assistance, Robotic Surgery
	• Treatment Planning
	• Drug Interaction Checks
Early Detection & Better Diagnosis	• Risk Factor Analysis (e.g., cancer)
	• Preventive Care (Kidney, heart disease etc.)
	• Pattern Recognition (e.g., Autism, Alzheimer and other little understood diseases)
Enhanced Administrative Efficiency	• Documentation and Workflow Automation
	• Mandatory Reporting to relevant Govt. Agencies
	• Billing Optimization
	• Communicating with Insurance Companies
Personalized Treatment	• Genetic Analysis (may be subject to local laws)
	• Custom Treatment Plans
	• Patient-Specific Approaches (e.g., combining with other non-medical holistic treatment)
New Drug Development	• Molecular Analysis
	• Clinical Trial Optimization
Remote/Virtual Patient Monitoring	• Wearable Integration (e.g., devices like Fitbeat)
	• Real-time Alerts (e.g., AI powered Holter monitor)
	• Chronic Disease Management of patients in rural areas
Greater Healthcare Access	• Telemedicine (particularly in developing countries)
	• Rural Healthcare
	• Healthcare Equity (for lower income patients)
Medical Training	• Virtual Simulations (for medical students, healthcare providers and medical doctors)
	• New Skill Development (for existing licensed practitioners)
	• Continuous Learning for medical professionals
Patient Education	• Interactive Learning
	• Health Literacy
	• Treatment Compliance (especially during clinical trial and for use of antibiotics)

4. Discussion

Over the past few decades, countries all over the world have been suffering from rising medical costs. Even governments in rich countries are facing challenges to solve this intractable problem. For example, in the US, over 20% of federal government spending goes to healthcare. Such rising costs is driven by multiple factors, including an aging population, increasing chronic disease burden, emergence of new diseases, administrative complexities, and the escalating costs of medical technologies and treatments. While there may not be an easy solution for tackling such a multifaceted problem, AI gives hope to significantly reduce healthcare costs while enhancing its affordability, efficiency and outcome for patients.

One of the less discussed topic in healthcare is the inequity with respect to how healthcare is delivered between countries and also within countries. While some continental European countries with good social safety nets pride themselves in offering universal health care, their systems are under stress (Mahishale, 2015). With rising longevity but declining population, which means a lower tax revenue to fund healthcare in the future, some drastic changes in the healthcare system are long overdue. It is plausible that many countries with sound universal healthcare system would follow the US footstep and opt for more outpatient services rather than inpatient services. In outpatient care, AI-enabled remote monitoring and virtual care platforms allow for more efficient patient management, reducing unnecessary hospital visits and stays while ensuring high standards of care. AI technologies therefore not only cut costs by preventing hospital readmissions and optimizing resource allocation, but also enhance patient satisfaction through more personalized and accessible healthcare services.

Aided by AI, developing countries that currently lack a robust healthcare infrastructure can quickly build new infrastructures at a fraction of the cost of developed countries. While some developed countries will have the first mover advantages in this area, developing countries can learn from the mistakes of developed countries such as the US and scale up AI applications to their healthcare sector. It should also be noted that a few developing countries such as Cuba, Jordan, Malaysia, Thailand to name a few, have made remarkable progress in developing a decent healthcare infrastructure. We also see pockets of healthcare excellence in different parts of India, although the country as a whole suffers from lack of proper healthcare for its population.

An added advantage of successful AI application to the healthcare sector of developing countries can ensure knowledge flow in both directions. With global warming, many tropical diseases that afflict large parts of Asia, Africa and South America will also become common in Europe and North America. However, knowledge of treating tropical diseases is limited in many western countries. For example, in the US, many medical doctors know more about Lyme disease, which affects a few thousand people in the US, more than they know about tuberculosis or malaria that afflict millions of people worldwide. Developing countries currently lack the wherewithal to develop effective treatments to such diseases. AI may help them bridge the knowledge gap, allow them to embark on advanced research, develop effective drugs and share the knowledge with the west, thereby creating a both-way traffic in medical knowledge creation and dissemination in place

of current system that is dominated by medical research and treatment developed in the West. The potential of collaboration between developed and developing countries is particularly significant in areas such as disease surveillance, drug development, and clinical research

While AI facilitated better healthcare at reduced costs would relieve budgetary pressure in many developed countries, it can also have many associated and multiplier effects in developing countries. For example, rather than relying on developed countries for advanced medical devices and equipment, developing countries can build their own medical device industry and use machine learning and other AI tools to manufacture medical equipment. Better medical treatment at reduced costs can also contribute to medical tourism in countries with rich history like Jordan. Professions such as nursing, allied health, palliative care as well as hospitality industry related to medical tourism can contribute to economic development of developing countries, in addition to reducing the health inequity that currently exists between developed and developing countries. AI can also be promising in reducing the rural-urban divide in healthcare in many countries.

5. Potentials of Jordan

Jordan's healthcare system can potentially be an important test case for replicating the AI applications from the US healthcare system and provide new insights as how to replicate and scale up such applications. Despite not being endowed with natural resources such as oil & gas like some of its neighbors, Jordan has developed a robust health care infrastructure for a country of its economic size. Jordan has a large pool of qualified medical doctors who are well-versed with US and western medical practices. Such pre-existing knowledge would make Jordan an ideal ground to experiment with various AI applications in its healthcare systems. The country's dual healthcare system, consisting of public and private healthcare facilities, requires adaptable AI solutions that can be implemented within the constraints of available resources. While private hospitals in urban areas like Amman and Irbid might afford to adopt sophisticated AI systems, healthcare facilities in smaller towns are likely to benefit more from adopting simplified AI applications focused on basic diagnostic support and patient management. Jordan's experience in managing refugee healthcare could provide valuable insights into how AI systems can be adapted for emergency response and resource-constrained settings (El Arab and Sagbakken, 2018). Possible emergence of Jordan as an AI augmented medical hub would also benefit the region of Middle East and North Africa.

6. Conclusion

While this paper presents an optimistic picture of the implications of AI applications to the healthcare sector, many challenges remain that can delay, if not derail, the application of AI to the healthcare sector. The rising nationalism across the globe can stymie cooperation across countries over the use of AI in medical care. The recent withdrawal of the US from the World Health Organization (WHO) and its imposition of export control on logic chips and other components used in making AI tools indicate that countries with advanced AI may attempt to keep a tight leash on the availability and use of various AI tools. The rivalry between US and China, the two superpowers of AI, may force other countries to choose sides and get embroiled in unnecessary conflicts. The emergence of Deep Seek and other AI platforms in China may upend the game and create both opportunities and challenges for other countries. Not unlike the digital divide, there is

a real danger of an ‘AI divide’ that may further exacerbate health inequity between countries rather than bridging this gap.

The politico-legal framework for application of AI to the healthcare sector is yet to be developed in most countries. This can significantly slow down the adoption of AI in the healthcare sector in many countries, thereby delaying the realization of full potential of AI augmented healthcare. In addition, the ethical implications of relying on AI to make crucial decisions (e.g., whether to keep or remove life support from a patient in vegetative state) is not yet clear to most experts and policy makers (Marques, Almeida and Pereira, 2024). Last but not the least, AI can result in significantly lower demand for medical doctors and other medical professionals and caregivers, thereby contributing to rising unemployment in different countries.

While much of the AI promises remain to be seen, this paper is offered as a prelude to further discussion on this topic so that we can avoid the mistakes that often accompany the introduction of new technologies and thereby proactively take steps to reduce the unintended consequences of the use of AI. The more we examine the use of AI in healthcare as well as in other fields that touch human lives, the better it will be for building a more empowered, inclusive, and just world.

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